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Introduction

Welcome to another course in the STEP series, **Siemens Technical Education Program**, designed to prepare our distributors to sell Siemens Industry, Inc. products more effectively. This course covers **Basics of Safety Switches** and related products.

Upon completion of **Basics of Safety Switches** you should be able to:

- Explain the need for circuit protection
- Identify fuse types and classes
- Explain the basic construction and operation of a Siemens safety switch
- Explain the operation and benefits of Siemens VBII safety switches and visible blade designs
- Identify various types of Siemens safety switches
- Explain the difference between fusible and non-fusible safety switches
- Identify circuit protection ratings for various types of Siemens safety switches
- Identify safety switch accessories

This knowledge will help you better understand customer applications. In addition, you will be better able to describe products to customers and determine important differences between products. You should complete **Basics of Electricity** before attempting **Basics of Safety Switches**. An understanding of many of the concepts covered in **Basics of Electricity** is required for **Basics of Safety Switches**.

After you have completed this course, if you wish to determine how well you have retained the information covered, you can complete a final exam online as described later in this course. If you pass the exam, you will be given the opportunity to print a certificate of completion from your computer.

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Safety Switches

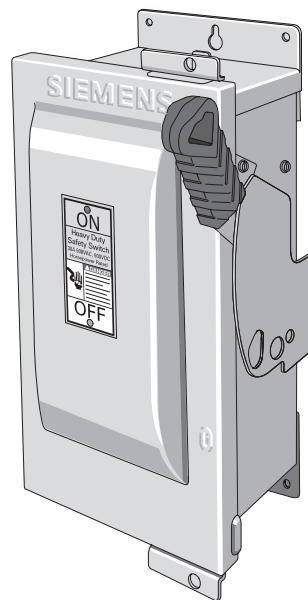
A **safety switch** is a common type of enclosed switch. Safety switches are generally used for two purposes:

- 1) As a disconnecting means for a service entrance
- 2) As a disconnecting means for motors

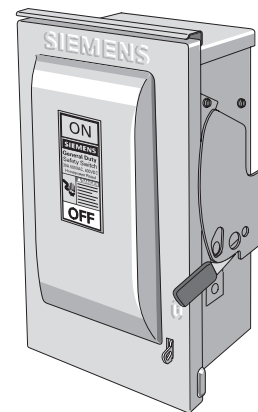
In either case, a safety switch may incorporate provisions for a fuse for overcurrent protection.

The safety switch enclosure provides a degree of protection to personnel against incidental contact with live electrical equipment. It also provides protection for the enclosed equipment against specific environmental conditions.

There are two families of Siemens safety switches: **general duty** and **heavy duty**.



Heavy Duty

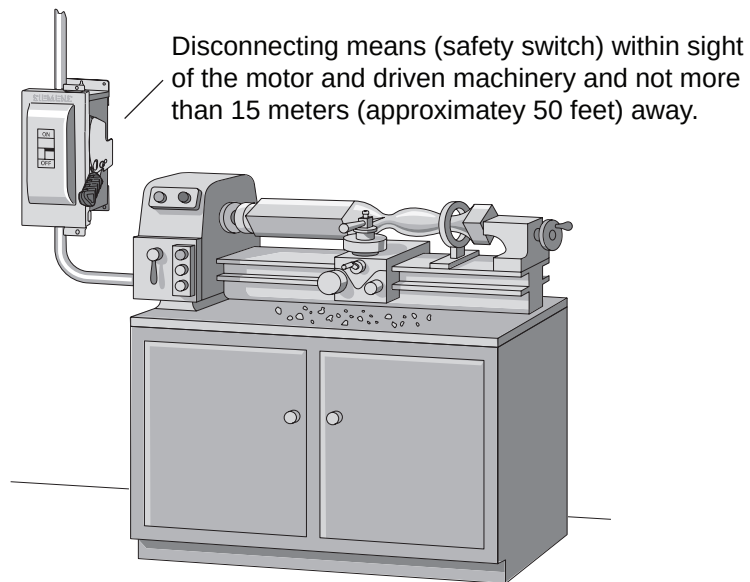


General Duty

Application

As previously indicated, a safety switch can be used as a disconnecting means for a motor. *National Electrical Code*® (*NEC*®) Article 430.102 requires a disconnecting means within sight of a motor and the machinery driven by the motor. The *NEC*® considers this to mean that the disconnecting device must be visible from the motor and machinery driven by the motor and not more than 15 meters (approximately 50 feet) away. *NEC*® Article 430.102 does include exceptions. Refer to the full article for details.

Regardless of where the safety switch is used, the function is to provide a means to connect and disconnect the load from its source of electrical power.



With power removed, someone can safely service the machinery without coming into contact with live electrical components or having the motor accidentally start.

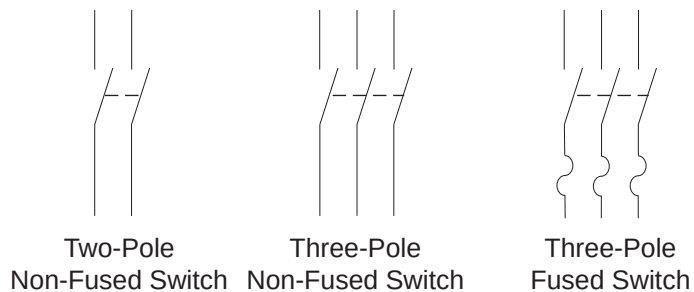
Additional Information

This course offers an introduction to safety switches, but more information is available on the Siemens Industry, Inc. web site. Among the booklets available are the **VB II Safety Switch Application and Selection Guide**, the **Safety Switch Cross-Reference Guide**, and the **Safety Switch Replacement Parts Guide**.

Symbols

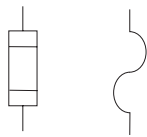
Switch Symbols

Symbols are used in a diagram to represent components. The symbols commonly used for a disconnect switch are shown below. The switch is normally shown in its "off" or "open" state.



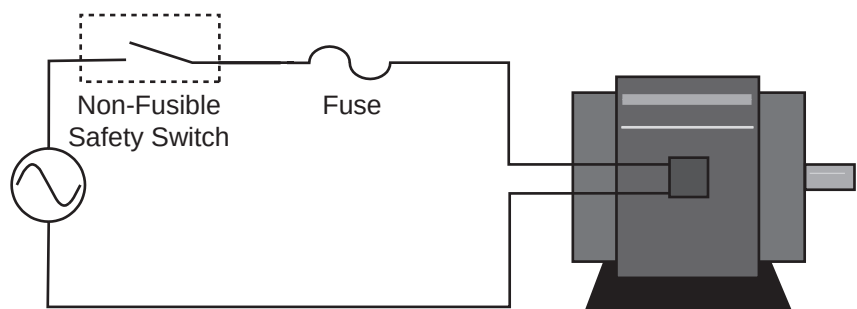
Fuse Symbols

Fuses are represented in an electrical circuit by either of the following symbols:



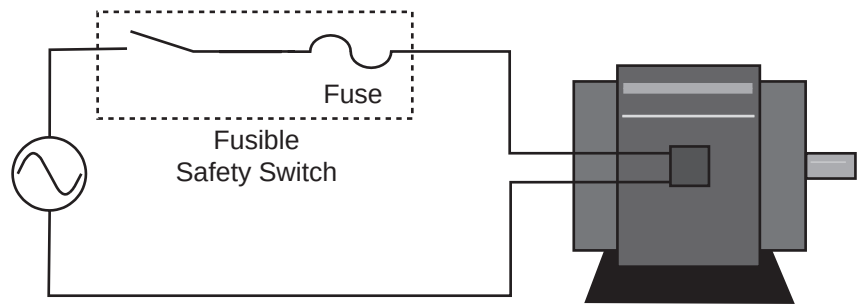
Non-Fusible Safety Switch

A safety switch with no associated fuses is referred to as a **non-fusible safety switch**. A non-fusible safety switch has no circuit protection capability. It simply provides a convenient means to open and close a circuit. Opening the circuit disconnects the load from its source of electrical power, and closing the circuit connects the load. Circuit protection must be provided by external overcurrent devices such as a circuit breaker or fuses. In the following illustration, power is supplied to a motor through a non-fusible safety switch and a separate fuse.



Fusible Safety Switch

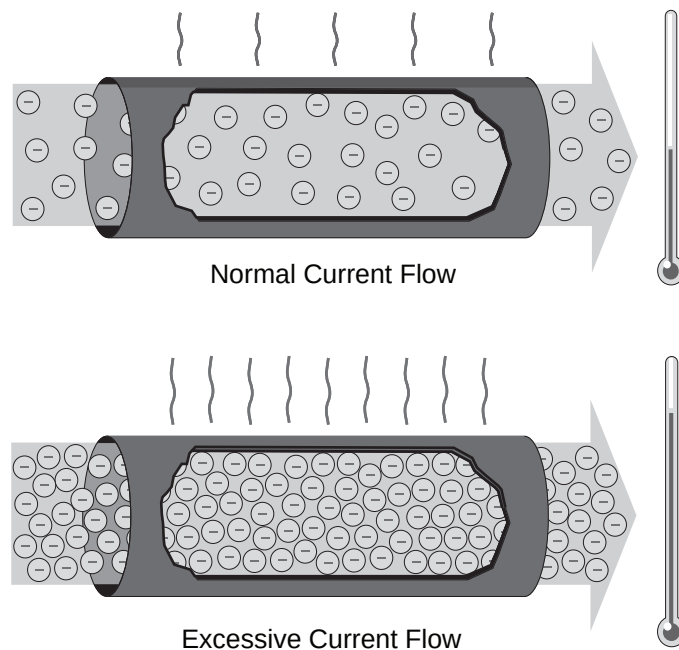
A safety switch can be combined with fuses in a single enclosure. This is referred to as a **fusible safety switch**. The switch provides a convenient means to manually open and close the circuit, and the fuse provides overcurrent protection.



Need for Circuit Protection

Current and Temperature

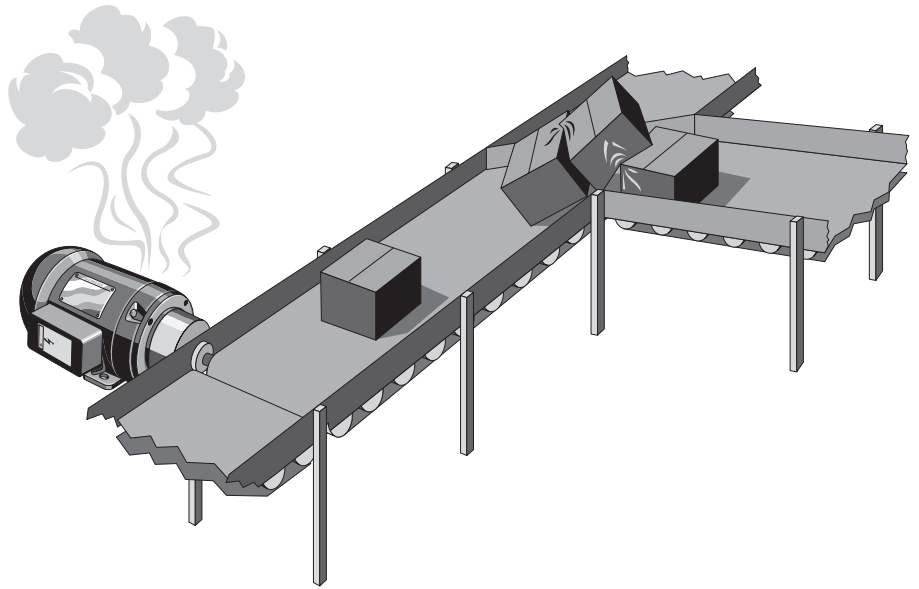
Current flow in a conductor always generates heat. The greater the current flow in a given size conductor, the hotter the conductor. Excess heat is damaging to electrical components and conductor insulation. For this reason conductors have a rated continuous current carrying capacity, or ampacity. Overcurrent protection devices, such as fuses, are used to protect conductors from excessive current flow. Fuses are designed to keep the flow of current in a circuit at a safe level to prevent the circuit conductors from overheating.



Excessive current is referred to as **overcurrent**. An overcurrent may result from a short circuit, overload, or ground fault. The first two types of overcurrent conditions are pertinent to this discussion.

Overloads

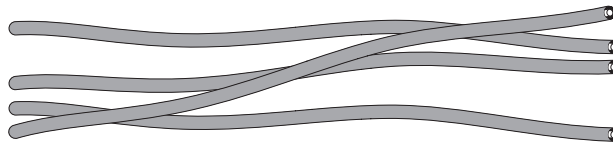
An **overload** occurs when too many devices are operated on a circuit or when electrical equipment is made to work beyond its ratings. For example, a motor rated for 10 amperes may draw 20, 30, or more amperes in an overload condition. In the following illustration, a package has become jammed on a conveyor, causing the motor to work harder and draw more current. Because the motor is drawing more current, it heats up. Damage will occur to the motor in a short time if the problem is not corrected or if the circuit is not shut down by the overcurrent protection device.



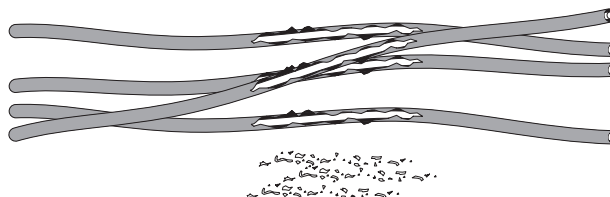
Conductor Insulation

Motors, of course, are not the only devices that require circuit protection for an overload condition. Every circuit requires some form of protection against overcurrent and the heat it produces. For example, high levels of heat can cause conductor insulation to break down and flake off, exposing the conductors.

Good Insulation

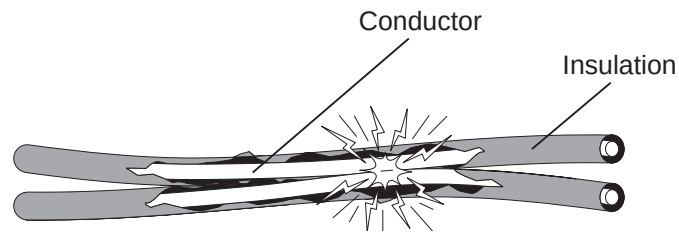


Insulation Damaged by Heat



Short Circuits

When exposed conductors touch, a **short circuit** occurs, and the circuit resistance drops to nearly zero. Because of this very low resistance, short-circuit current can be thousands of times higher than normal operating current.



Ohm's Law shows how current, voltage, and resistance are related. For example, a 240 volt motor with 24 ohms of resistance would normally draw 10 amperes of current.

$$I = \frac{E}{R}$$

$$I = \frac{240}{24}$$

$$I = 10 \text{ A}$$

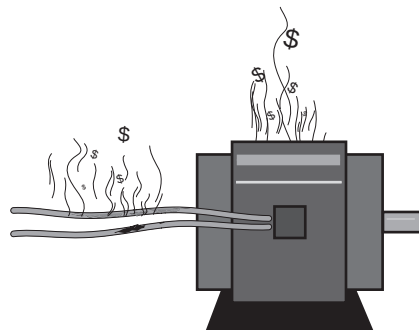
When a short circuit occurs, resistance drops dramatically. For example, if the above resistance dropped to 0.024 ohms due to a short circuit, the current would increase to 10,000 amperes.

$$I = \frac{240}{0.024}$$

$$I = 10,000 \text{ A}$$

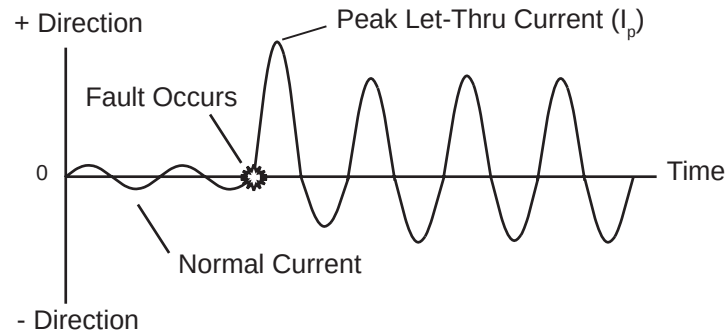
Preventing Damage

The heat generated by short-circuit current can rise to dangerous levels quickly, causing extensive damage to conductors and connected equipment. This means that current must be interrupted instantaneously when a short circuit occurs. Slight overcurrents can be allowed to continue for some period of time, but, as the overcurrent magnitude increases, the protection device must act more quickly.

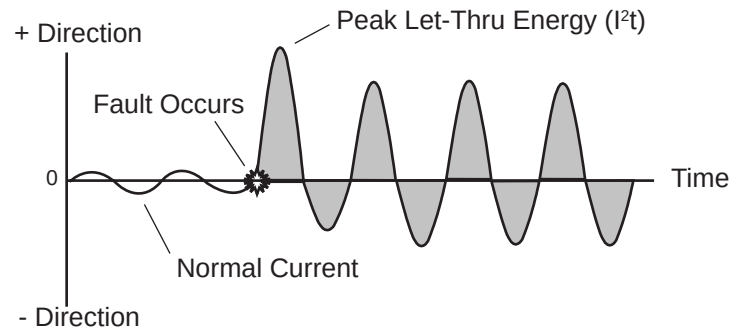


Short-Circuit Current in Unprotected Electrical Circuits

When a short circuit occurs in an unprotected circuit, current will continue to flow until the circuit is damaged or the power is removed manually. The peak short-circuit current of the first cycle is the greatest and is referred to as **peak let-through current (I_p)**. In addition to the damage associated with heat, the electromagnetic force associated with this current can cause mechanical damage to electrical components.

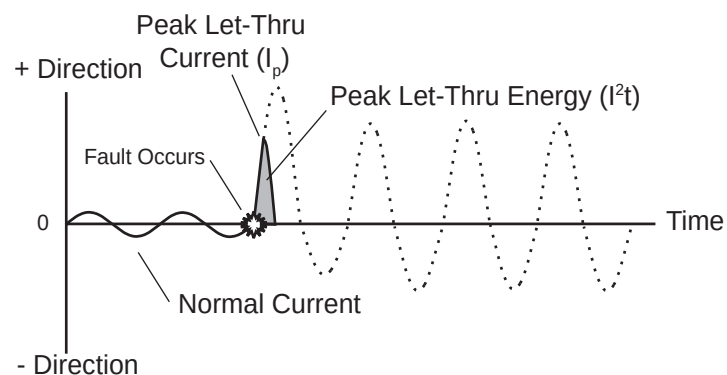


The **maximum destructive energy let-through (I^2t)** is a measure of the energy associated with this current. It is capable of producing enough heat to melt conductors.



Short-Circuit Current in Protected Electrical Circuits

A properly applied overcurrent protection device will open the circuit quickly when a short circuit occurs, limiting peak let-through current (I_p) and energy (I^2t).



NEC® Articles

Various *NEC*® articles discuss overcurrent protection. Some of the articles relevant to this topic are listed below.

- Article 210 covers branch circuits
- Article 215 covers feeders supplying branch circuit loads
- Article 230 covers service conductors and their control and protection
- Article 240 covers general requirements for overcurrent protection and overcurrent protection devices (up to 600 volts)
- Article 430 covers motors, motor circuits, and controllers

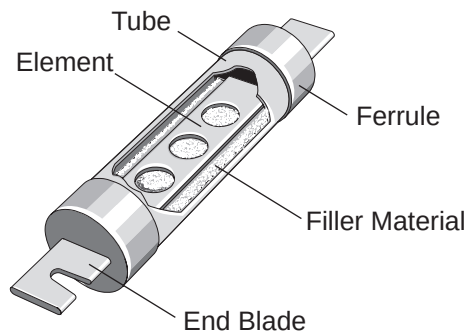
It is beyond the scope of this course to cover this content; but it is useful to consider the intent of this information. In general, these articles are designed to ensure that conductors and overcurrent protection devices are properly sized for their loads and that overcurrent protection devices provide the appropriate level of protection for conductors in the event of an overcurrent.

Fuses

Circuit protection would be unnecessary if overloads and short circuits could be eliminated. Unfortunately, they do occur. To protect a circuit against these destructive currents, a protective device automatically disconnects the electrical equipment from the power source when a fault condition occurs. A **fuse** is the simplest device for interrupting a circuit experiencing an overload or a short circuit.

Fuse Construction

A typical fuse, like the one shown below, consists of an element connected to ferrules. These ferrules may also have attached end blades. The element provides a current path through the fuse. It is enclosed in a tube and surrounded by a filler material.



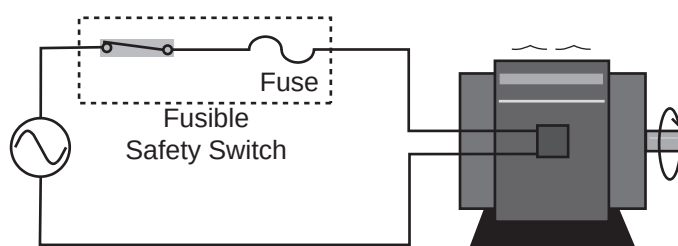
Closed Switch Symbol

As mentioned earlier, switches are normally shown in their "off" or "open" position. For the purpose of illustration, the following symbol can be used to show a switch closed, connecting the load to the power source. This is not a legitimate symbol. It is used here for illustrative purposes only.



Using a Fuse in a Circuit

In the following example, a motor is connected to a voltage source through a fusible safety switch. The switch and fuse function as part of the conductor supplying power to the motor.



Fuse Subject to Overcurrent

Current flowing through the fuse element generates heat, which is absorbed and dissipated by the filler material. When an overcurrent occurs, temperature in the element rises. In the event of a transient overload condition, the excess heat is absorbed by the filler material. However, if a sustained overload occurs, the heat will eventually melt open an element segment. This will stop the flow of current.

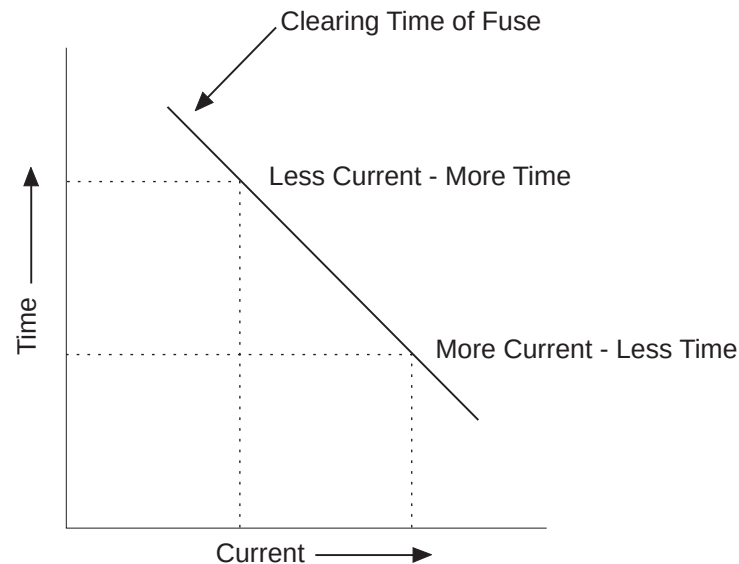


Fuse During Fault

Fuse After Fault

Fuse Clearing Time

Fuses have an **inverse time-current characteristic**. The greater the overcurrent, the less time it takes for the fuse to open. This is referred to as the **clearing time** of the fuse.



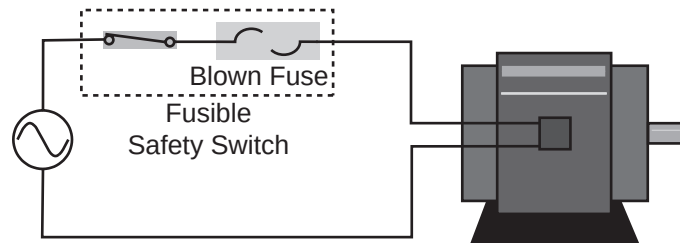
Open Fuse Symbol

For the purpose of explanation, the following symbol is used to show an open fuse, commonly referred to as a blown fuse. This is not a legitimate symbol. It is used here for illustrative purposes only.



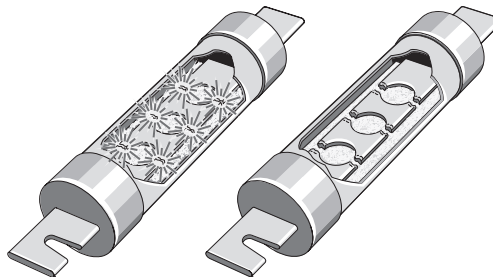
Overload Current

In the following example of a motor circuit, an overload has occurred, causing the fuse to open and remove power from the motor. As a result, the motor is stopped even though the switch is closed. Keep in mind that the fuse in the fusible safety switch is sized to protect the conductors that supply current to the motor. Overload protection for the motor is normally provided separately, often by an overload relay.



Short-Circuit Current

Short-circuit current, which can be several thousand amperes, generates extreme heat. When a short circuit occurs, several element segments melt simultaneously, quickly disconnecting the load from the power source. Short-circuit current is typically cut off in less than half a cycle, before it can reach its full value.



Fuse During Fault Fuse After Fault

Non-time-Delay Fuses

Non-time-delay fuses, also called **fast-acting fuses**, provide excellent short-circuit protection. Non-time-delay fuses usually hold 500% of their rating for approximately one-fourth of a second, after which the current-carrying element melts. This means that these fuses should not be used in motor circuits, which often have starting currents greater than 500% and lasting longer than one-fourth of a second.

Time-Delay Fuses

Time-delay fuses provide both overload and short-circuit protection. Time-delay fuses, when properly sized, allow a higher starting current for a sufficient time to allow a motor to start without a nuisance opening of the fuse. However, a longer lasting overload will cause the fuse to open.

Fuse Ratings and Classifications

Ampere Rating

Each fuse has a specific **ampere rating**, which is its continuous current-carrying capacity. The ampere rating of the fuse chosen for a circuit usually should not exceed the current-carrying capacity of the circuit. For example, if a circuit's conductors are rated for 10 amperes, the largest fuse that should be selected is 10 amperes.

However, there are circumstances where the ampere rating is permitted to be greater than the continuous current-carrying capacity of the circuit. For example, motor and welder circuit fuse ratings can exceed conductor ampacity to allow for inrush currents and duty cycles within limits established by the *NEC*[®].

Voltage Rating

The **voltage rating** of a fuse must be at least equal to the circuit voltage. The voltage rating of a fuse can be higher than the circuit voltage, but never lower. A 600 volt fuse, for example, could be used in a 480 volt circuit, but a 250 volt fuse could not be used in a 480 volt circuit.

Interrupting Rating

Fuses are also rated according to the level of fault current they can interrupt. This is referred to as the **interrupting rating** of the fuse and is expressed in amperes (often shortened to amps). A fuse for a specific application should be selected so that it can sustain the largest potential short-circuit current that could occur in the application. This is important because, if the fault current exceeded the interrupting ability of the fuse, the fuse could rupture and extensive damage could occur.

Fuse Classes

Underwriters Laboratories (UL) establishes and standardizes basic performance and physical specifications in developing its safety test procedures. These specifications have resulted in distinct classes of low voltage fuses (600 volts or less). The following chart lists selected UL fuse classes.

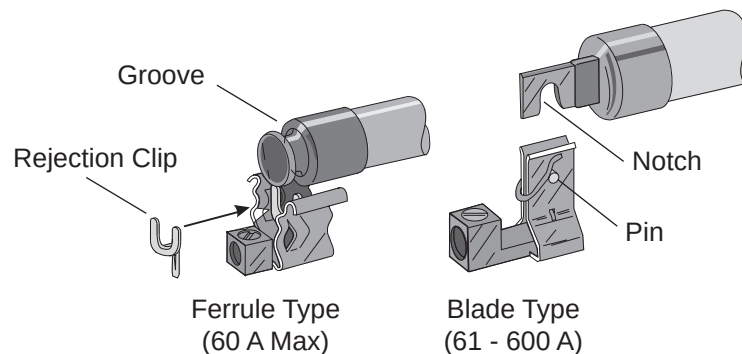
Fuse Class	Fuse Overload Characteristic	Ampere Ratings	AC Voltage Ratings	Interrupting Rating
H	Renewable Fuses, Fast-acting	1-600 A	250 V, 600 V	10,000 A
K5	Fast-acting	1-600 A	250 V, 600 V	50,000 A
J	Time-delay	1-600 A	600 V	200,000 A
J	Fast-acting	1-600 A	600 V	200,000 A
RK1	Time-delay	0.1-600 A	250 V, 600 V	200,000 A
RK1	Fast-acting	1-600 A	250 V, 600 V	200,000 A
RK5	Time-delay	0.1-600 A	250 V, 600 V	200,000 A
T	Fast-acting	1-1200 A	300 V, 600 V	200,000 A
L	Time-delay	200-6000 A	600 V	200,000 A

Current Limiting Fuses

Fuses are also grouped into **current limiting** and **non-current limiting** categories based on their operating and construction characteristics. When a short circuit occurs, current limiting fuses are designed to open more quickly than non-current limiting fuses, significantly limiting peak let-thru current and peak let-thru energy.

For example, all Class R fuses, even those marked as time-delay fuses, are current limiting. The time delay applies to overloads and not short circuits.

In addition, Class R fuses incorporate rejection clips or pins that permit only class R fuses to be installed. This prevents installation of a fuse with a lower interrupting rating, such as a Class H or K fuse.



Class R fuses are not the only current limiting fuses. Consult appropriate UL and fuse manufacturer literature for additional information on fuses.

Enclosures

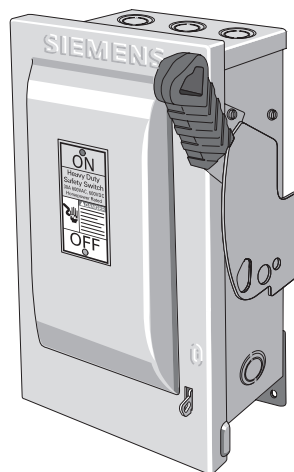
For the purpose of this course, an enclosure is the case that houses the components of an electrical device. The function of the enclosure is to prevent someone from accidentally touching an internal component that may have voltage applied and to protect internal components from damage.

Various standards describe enclosure types. One of the more frequently cited standards is NEMA standard 250. In addition to NEMA standard 250, published by National Electrical Manufacturers Association, UL 50 and UL 508, published by Underwriters Laboratories Inc., are also important standards for electrical equipment enclosures. These standards provide enclosure descriptions, features, and test criteria for hazardous and nonhazardous locations.

The following brief descriptions cover enclosures available for Siemens safety switches. Within the industry, it is common to refer to the enclosure type numbers as NEMA types, but these type numbers also apply to UL 50 and UL 508.

Type 1 Enclosures

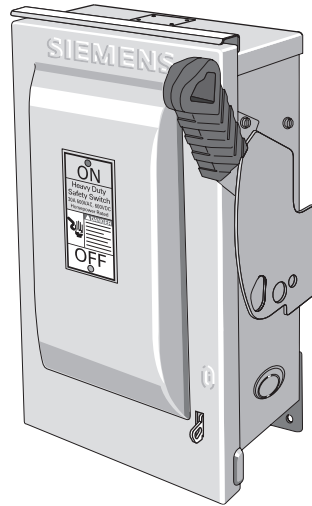
Type 1 enclosures are intended for indoor use primarily to provide protection against limited amounts of falling dirt and contact with the enclosed equipment in locations where unusual service conditions do not exist.



Type 1 Enclosure

Type 3R Enclosures

Type 3R enclosures are intended for outdoor use primarily to provide a degree of protection against falling rain and sleet. They are not intended to provide protection against conditions such as dust, internal condensation, or internal icing.

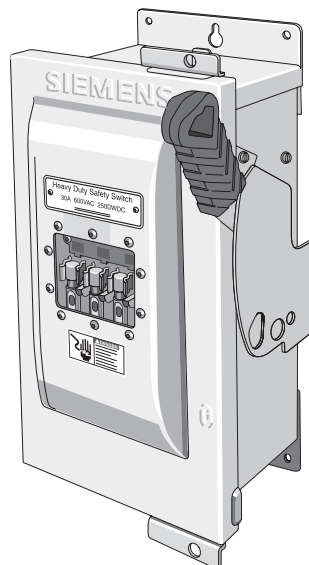


Type 3R Enclosure

Types 4 and 4X Enclosures

Type 4 enclosures are intended for indoor or outdoor use primarily to provide a degree of protection against windblown dust, rain, splashing water, hose-directed water, and damage from external ice formations. They are not intended to provide protection against conditions such as internal condensation or internal icing.

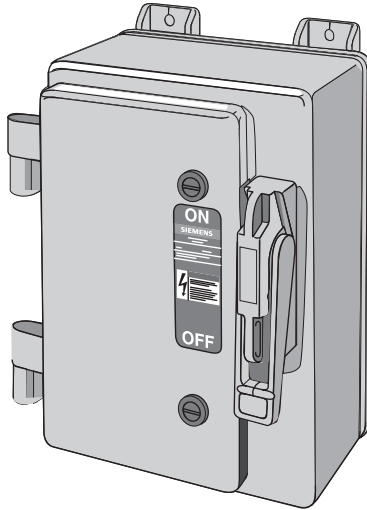
Type 4X enclosures are made of a material such as stainless steel and are intended primarily to provide a high degree of protection against corrosion, windblown dust and rain, splashing water, and hose-directed water.



Type 4/4X Stainless Enclosure
with Viewing Window

Non-metallic 4X Enclosures

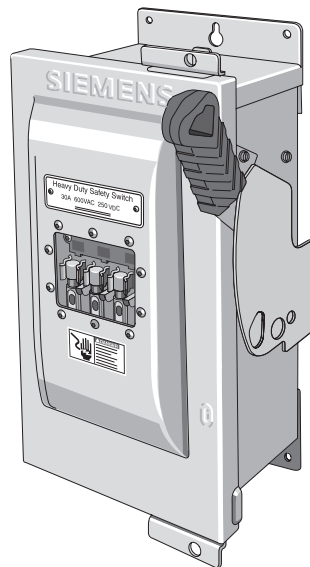
Another safety switch enclosure is a fiberglass-reinforced polyester version of the 4X enclosure. This **non-metallic 4X enclosure** has no external metal parts.



Type 4X Non-Metallic Enclosure

Types 12 Enclosures

Type 12 enclosures provide a degree of protection against dust, falling dirt, and dripping water in indoor locations, but are not intended to protect against conditions such as internal condensation.

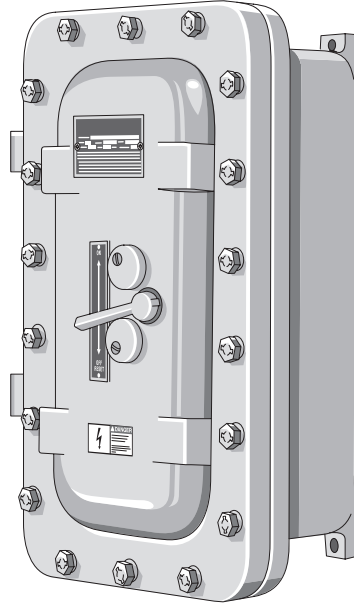


Type 12 Enclosure
with Viewing Window

Types 7 and 9 Enclosures

Type 7 enclosures are intended for indoor use in locations classified as Class I, Groups A, B, C, or D, as defined in the *NEC*®.

Type 9 enclosures are intended for indoor use in locations classified as Class II, Groups E, F, or G, as defined in the *NEC*®.



Type 7 and 9 Enclosure

Hazardous Environments

Articles 500 through 504 of the *NEC*® cover the use of electrical equipment in locations where fire or explosions due to gas, flammable liquids, combustible dust, or ignitable fibers may be possible. While you should never specify a hazardous location, it is important to understand the regulations that apply. It is the user's responsibility to contact local regulatory agencies to define the location as Division I or II and to comply with all applicable codes.

Divisions

Division I refers to a situation where hazardous materials are normally present in the atmosphere. **Division II** identifies conditions where the atmosphere may become hazardous as a result of abnormal conditions. For example, if a pipe carrying a hazardous material developed a leak, the surrounding atmosphere could become hazardous.

Classes and Groups

Hazardous locations are further identified by class and group. **Class I, Groups A, B, C, and D** are chemical gases or liquids.

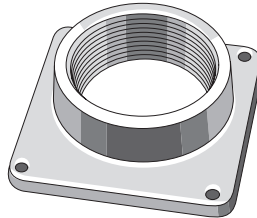
Class II, Groups E, F, and G include flammable dust.

Class III includes all ignitable fibers and lints, such as clothing fiber in textile mills, and flyings, such as saw dust. Class III is not divided into groups.

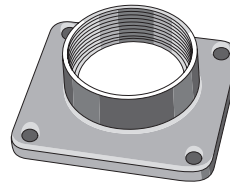
	Class I		Class II		Class III
	Groups A-D Gases and Liquids		Groups E-G Flammable Dust		
A	Acetylene	E	Metallic Dust		Ignitable Fibers
B	Hydrogen	F	Carbon Dust		Flyings
C	Acetaldehyde Ethylene	G	Grain, Plastic or Chemical Dusts		
D	Acetone Gasoline Methanol				

Hubs

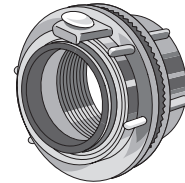
Various **hubs** are available for attaching cable conduit to the enclosures.



ECHV300
3" Conduit Hub
Type 3R Enclosure



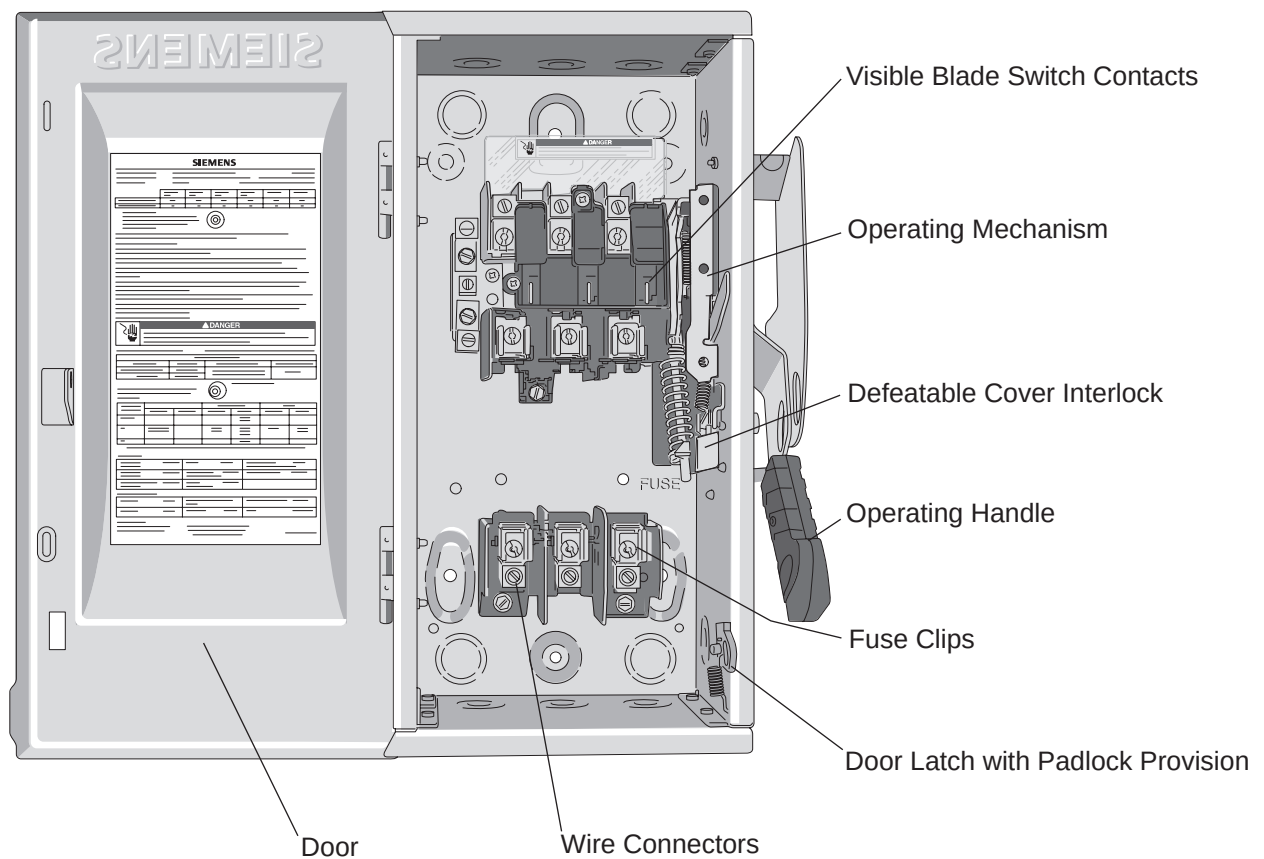
ECHS200
2" Conduit Hub
Type 3R Enclosure



SSH150
1.5" Conduit Hub
Type 4/4X Enclosure

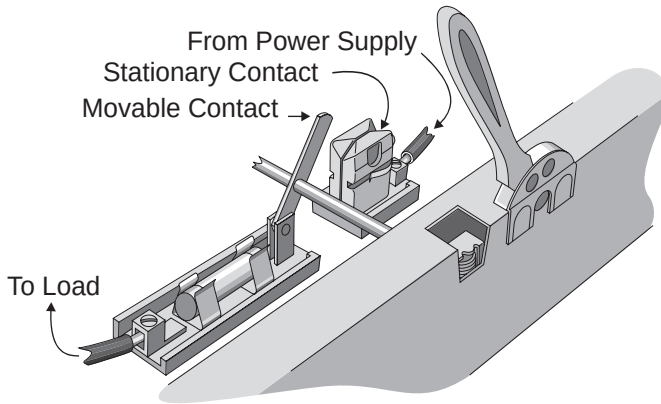
Switch Design

The enclosure houses the switch mechanism, wire connectors, and an operating mechanism. A handle, connected to the operating mechanism, opens and closes the visible blade contacts. If the switch is fusible, the enclosure also houses the fuse clips. Provisions for locking the door and/or switch handle are provided.

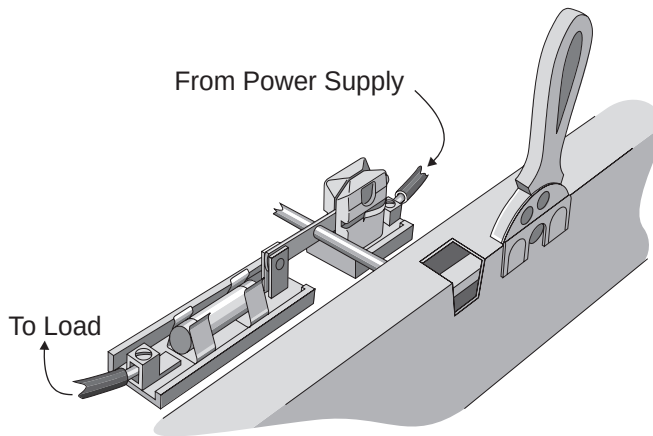


Knife Blade Switch Principle

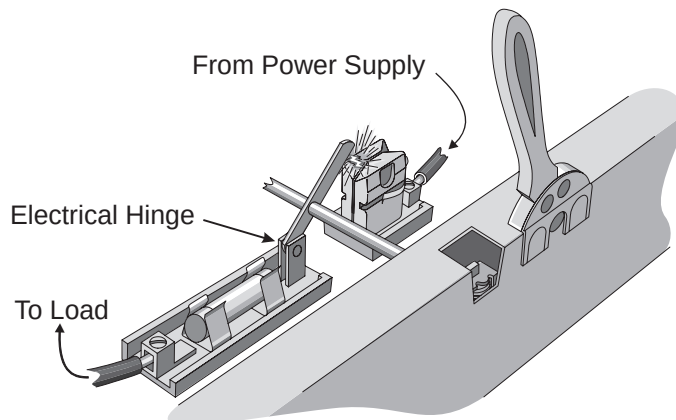
Switches use contacts to break the circuit and stop the flow of current. A typical switch assembly consists of a stationary contact, a hinged movable contact, and an operating handle. The hinged movable contact may also be referred to as a **knife blade**. If the movable contact is not touching the stationary contact, no current flows.



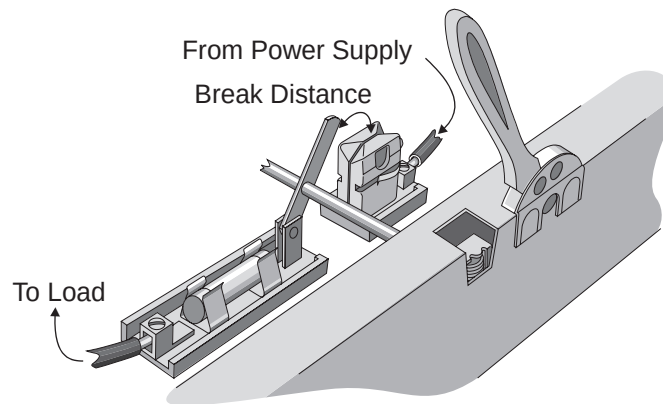
Moving the handle to the "on" position closes the contacts and provides a complete path for current to flow from the power supply to the load.



Moving the handle to the "off" position opens the contacts, interrupting the flow of electricity. As the contacts start to open, current continues to flow across the air gap between the two contacts in the form of an arc. Current continues to flow until the physical distance between the contacts is great enough to interrupt the flow of current.



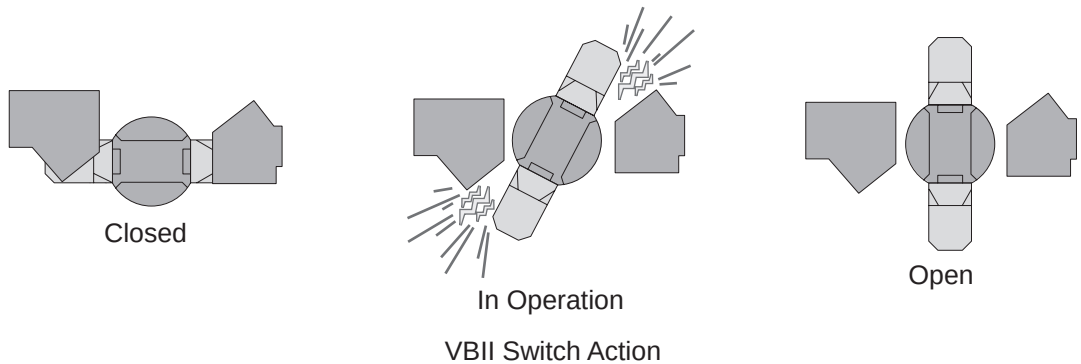
The point at which the arc is extinguished is called the **break distance**.



VBII Safety Switch Design

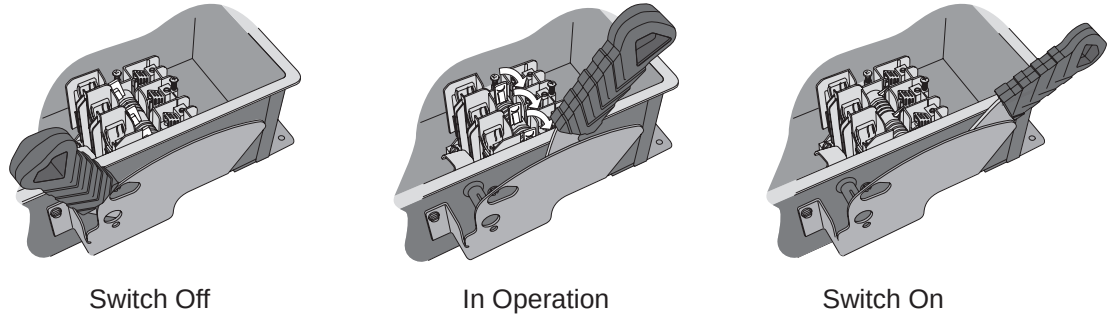
Unlike the knife-blade switch, the switching action of the Siemens 30-200 A **VBII Safety Switch** breaks the arc in two places. As a result, two smaller arcs are created, and heat generation is reduced. The switching speed is also increased, since the breaking distance is effectively doubled. The overall result is enhanced performance and increased longevity.

Also, in contrast to the knife blade switch, the VBII Safety Switch blades are self-aligning, ensuring positive contact. Furthermore, the electrical hinge, a wear and friction point, has been eliminated. The result is a fast, positive, and reliable switching action.



Over-center-toggle Switch Action

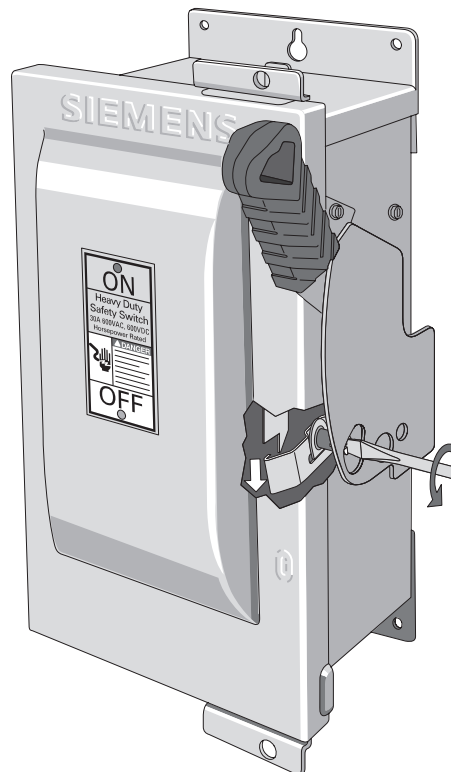
Another feature which enhances the speed of switching is the **over-center-toggle** design. During operation of the switch, as the handle is moved past the midpoint, the switch suddenly and rapidly snaps from off to on or from on to off, depending upon the direction of movement of the handle. Besides enhancing the switching speed, this also gives a positive feel to the switch operation.



VBII "Over-Center-Toggle" Action

Defeatable Cover Interlock

The VBII **cover interlock** prevents someone from opening the door while the switch is in the "on" position. Normally, the interlock also prevents someone from turning the switch on with the door open. However, for the purposes of testing or servicing, the door interlock is defeatable. As shown in the following illustration, this can be done with an ordinary screwdriver.



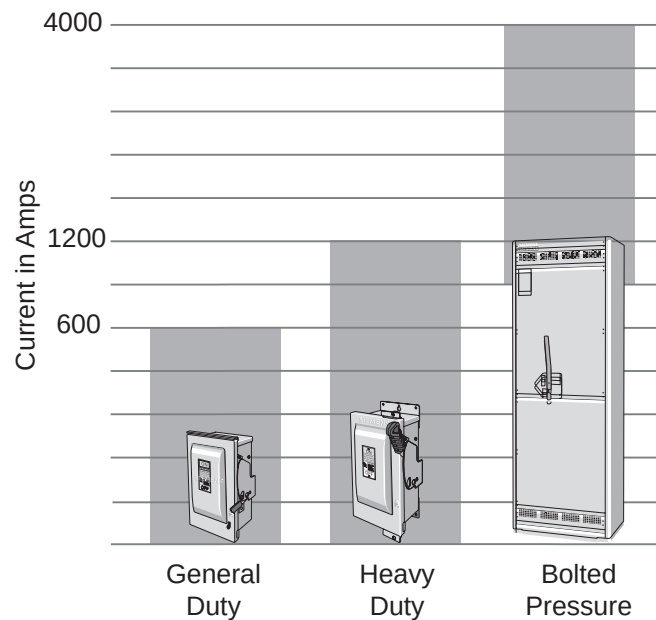
Safety Switch Ratings

Ampere Rating

Every safety switch has a **current rating**, also called an **ampere rating**, which is the maximum continuous current the switch is designed to carry.

For example, Siemens VBII general duty switches are available with ampere ratings of 30, 60, 100, 200, 400, and 600 amperes. Siemens VBII heavy duty switches are rated for 30, 60, 100, 200, 400, 600, 800, and 1200 amperes.

When higher ampere ratings are required, a bolted pressure switch can be used. A bolted pressure switch is designed so that a high clamping pressure is placed on all blade joints. Though not covered in this course, Siemens bolted pressure switches are available with ratings of 800, 1200, 1600, 2000, 2500, 3000, and 4000 amperes.



Short-circuit Current Withstand Rating

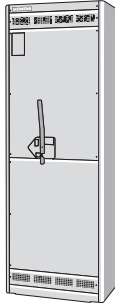
The maximum short-circuit current that a safety switch can carry for a short time is called its **short-circuit current withstand rating**.

For example, Siemens VBII general duty switches have a maximum short-circuit current withstand rating of 100,000 amperes, while the maximum short-circuit withstand rating of Siemens VBII heavy duty switches is 200,000 amperes. The short-circuit withstand rating for a specific switch depends on the fuse class used.

Voltage Rating

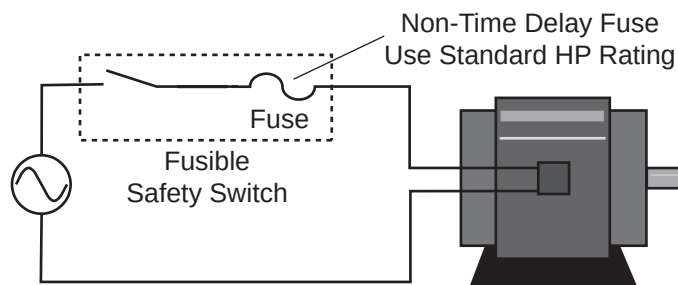
Safety switches are also rated according to the maximum voltage they can handle. The **voltage rating** of the switch must be at least equal to the circuit voltage. In other words, it can be higher than the circuit voltage, but never lower. For example, a safety switch rated for 600 volts can be used on a 480 volt circuit, but a switch rated for 240 volts must not be used on a 480 volt circuit.

The following chart shows the available voltage ratings for Siemens safety switches and bolted pressure switches.

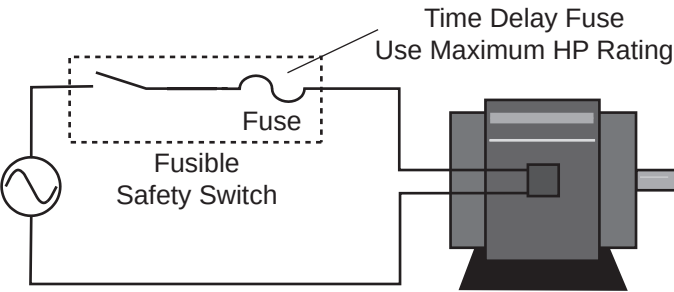
		
General Duty	Heavy Duty	Bolted Pressure
240 VAC	240 VAC	240 VAC
250 VDC	600 VAC	480 VAC
	600 VDC	600 VAC*
		*600 VAC Bolted Pressure Switch is not UL Listed

Dual Horsepower Ratings

All Siemens safety switches are as **dual horsepower rated**, which means that they have two horsepower ratings for motor applications. For example, a switch might have a standard rating of 10 HP and a maximum rating of 30 HP. The standard rating of 10 HP applies when non-time delay fuses are used.



The maximum rating of 30 HP applies when time delay fuses are used.



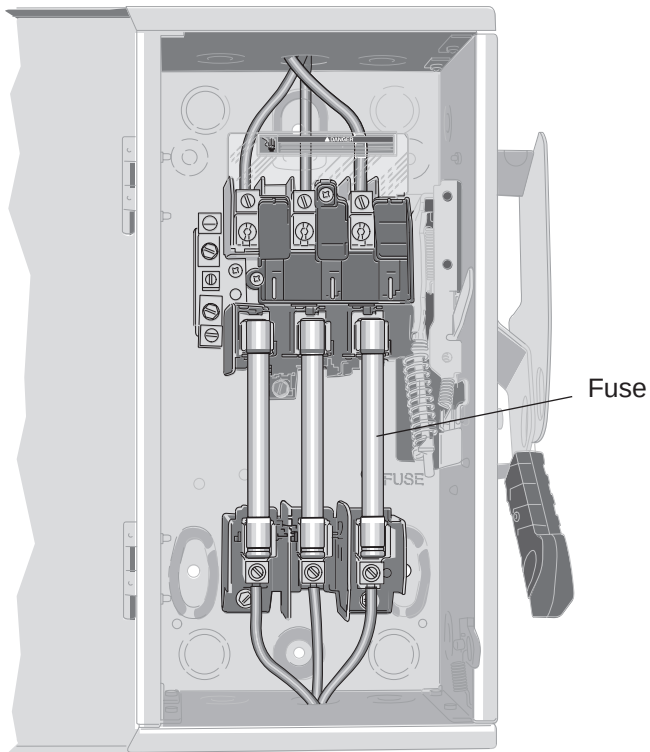
The following chart reflects the range of horsepower ratings for Siemens safety switches. Refer to the Speedfax catalog for the standard and maximum horsepower ratings for specific catalog numbers.

Safety Switch Type	Voltage	Horsepower Range
General Duty	240 VAC	1½-200
	250 VDC	5-50
Heavy Duty	240 VAC	1½-250
	600 VAC	3-500
	250 VDC	5-50
	600 VDC	15-50

Switch Circuit Types and Terminology

Pole

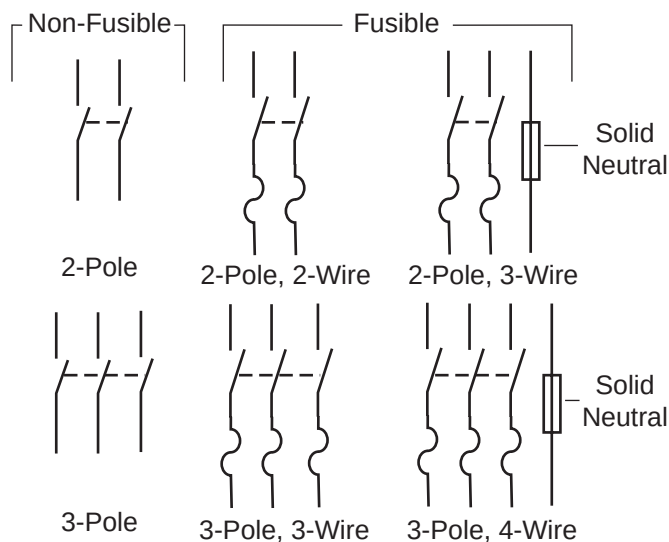
The term **pole** refers to the number of circuits that can pass through a switch at one time. This is the number of circuits that the device can connect and disconnect. The following drawing, for example, shows a 3-pole safety switch. The three circuits are mechanically connected so that all three poles connect and disconnect the line and load simultaneously when the switch is operated. In this example, each pole is fused for overcurrent protection.



3-Pole Fusible Safety Switch

Circuit Configurations

Circuit configuration diagrams for 2-pole and 3-pole safety switches are shown below. Safety switches may be non-fusible, fusible, or fusible with a solid neutral.

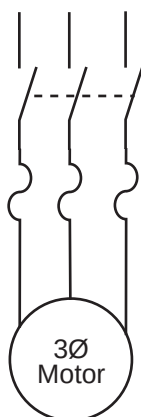


Siemens safety switches are available in all the configurations shown above as well as additional configurations. For example, Siemens heavy duty fusible and non-fusible safety switches are also available with four or six poles.

Example

The circuit configuration required depends on the load and the power supply connected to it. For example, a 3-phase motor needs a 3-pole switch to connect it to a 3-phase power supply. If overcurrent protection is required, a fusible 3-pole safety switch should be selected, as in the following example.

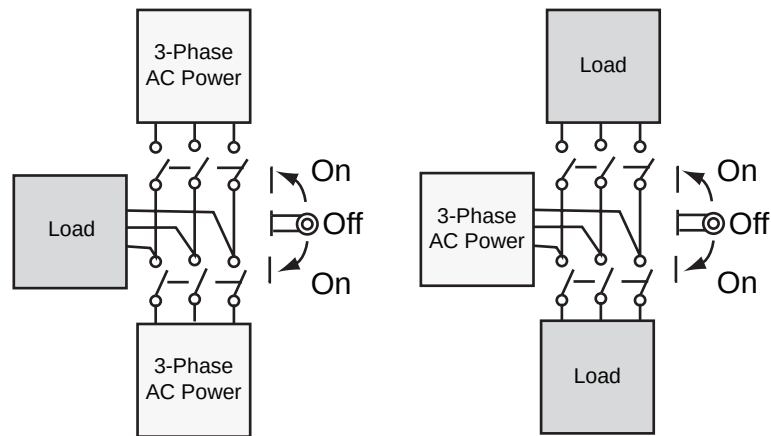
3Ø AC Power Supply



Throw

Throw is the term used to specify the number of circuits to which a conductor can be connected. All the examples shown so far have been for single throw switches. However, Siemens also offers double throw switches in both general duty non-fusible and heavy duty fusible and non-fusible designs.

Double throw switches are intended to transfer loads from one power source to another or to connect a single power source to either of two loads. For example, the illustration shown below shows 3-pole, non-fusible double throw switches. For either of the two applications, no power is applied to a load with the switch in the center (off) position, and only one set of contacts can be closed at a time.



In the application on the left, with the switch in the up position, the upper power source is connected to the load. With the switch in the down position, the lower power source is connected to the load.

In the application on the right, with the switch in the up position, the power source is connected to the upper load. With the switch in the down position, the power source is connected to the lower load.

VBII General Duty Safety Switches

VBII general duty switches are intended for use primarily on power supplies rated at 240 VAC or less, where the available fault current is less than 100,000 amperes (with Class R or T fuses, or 10,000 A max with Class H fuses). They can be supplied in a Type 1 (indoor) or Type 3R (outdoor) enclosure.

Plug Fuse Type Safety Switch

The **general duty plug fuse type switch** is available for 120 volt or 240 volt systems. It is suitable for 1-pole or 2-pole applications, and is rated at 30 amperes. A separately supplied, 30 ampere Type S plug fuse is required. This switch is available for use on 2-wire or 3-wire motor applications up to three horsepower. A non-fusible model comes in a 2-pole configuration. It is rated at 60 amperes and can be used with motors up to 10 HP. Pullout models are also available in fused and non-fused versions.

General Duty Switches

Fusible general-duty safety switches are available with two or three poles (both with a solid neutral) or with four poles. Fusible switches accept Class H fuses as standard. A field-installable rejection kit is available which rejects all but Class R fuses.

Non-fusible general duty safety switches are available with two or three poles. All general duty switches have both cover and handle padlocking capabilities.

Ratings

Ampere ratings: 30, 60, 100, 200, 400, or 600 A

Fuses: Class H, Class K, Class R (Class R fuse clip rejecter kit required), Class T (200 to 600 A switches, 200 A switches require field adapter kit)

Voltage ratings: 240 VAC/250 VDC

Short-circuit current withstand ratings: Suitable for use on systems capable of delivering not more than 100,000 RMS symmetrical amperes of fault current when Class R fuses are installed. 200 to 600 ampere switches with Class J and T fuses also rated for use in circuits with potential fault current up to 100,000 RMS symmetrical amperes.

Enclosures

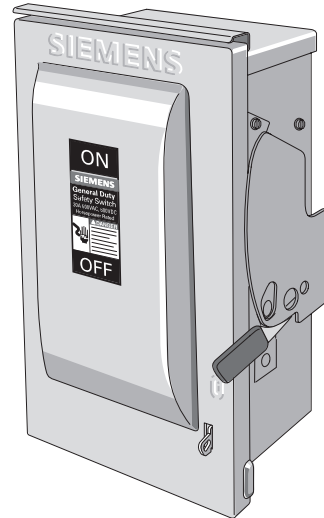
General duty switches are available with a **Type 1 enclosure**, which is intended for indoor use. These switches have interlocks to prevent the cover from being opened when the switch is in the "on" position and to prevent the switch from being turned on with the door open. (There is a front-operable release for this feature.)

This enclosure is intended primarily to provide protection against contact with the safety switch and is used in locations where unusual service conditions do not exist.

General duty 2-pole and 3-pole safety switches are also available with a **Type 3R enclosure**, which is intended for outdoor use and provides a degree of protection against falling rain and sleet. It is also able to withstand the formation of ice on the enclosure without damage, but is not intended to provide protection against conditions such as dust, internal condensation, or internal icing.



General Duty
Safety Switch
Type 1 Enclosure



General Duty
Safety Switch
Type 3R Enclosure

VBII Heavy Duty Safety Switches

Ratings

VBII heavy duty safety switches can be used on power supplies up to 600 Volts, AC or DC, in applications where the available fault current is 200,000 amperes or less. Interlocks prevent someone from inadvertently opening the cover while the switch is in the "on" position or inadvertently turning on the switch while the cover is open. Heavy duty safety switches also have cover and handle padlocking capabilities.

Ampere ratings: 30, 60, 100, 200, 400, 600, 800, or 1200 A

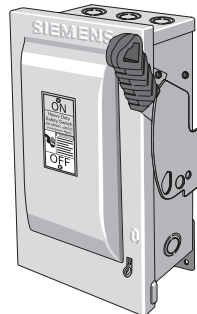
Fuses: Class H, Class K, Class R (Class R fuse clip rejecter kit are required), Class J (240 and 600 V switches, 600 V switches are field convertible) Class L (800 and 1200 A switches only), Class T (100 to 1200 A switches, 100 A and 200 A switches require an adapter kit)

Voltage ratings: 240/480/600 VAC; 250/600 VDC

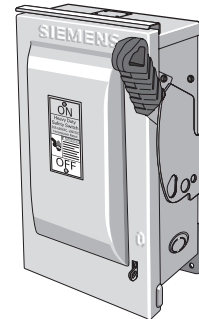
Short-circuit current withstand ratings: Suitable for use on systems capable of delivering not more than 200,000 RMS symmetrical amperes of fault current when Class J or R fuses are installed, except the 800 and 1200 ampere switches, which are suitable for use on circuits capable of delivering not more than 200,000 RMS symmetrical amperes of fault current when Class L fuses are installed. 100 to 1200 ampere switches with Class T fuses and field adapter kit are also rated for use in circuits with potential fault current up to 200,000 RMS symmetrical amperes.

Enclosures

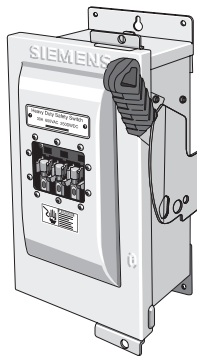
Siemens offers a broad selection of heavy duty safety switches with Type 1 or 3R enclosures. Selected heavy duty safety switches are also available with other enclosure types such as Types 4/4X stainless steel with viewing window, Type 4X non-metallic, and Type 12 with viewing window.



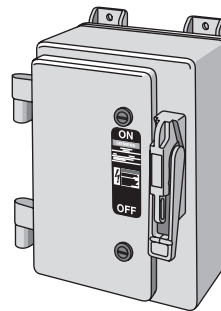
Type 1



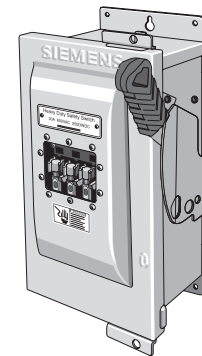
Type 3R



Type 4/4X Stainless
with Viewing Window



Non-Metallic Type 4X



Type 12 with
Viewing Window

Siemens safety switches with Type 4/4X stainless steel or Type 12 enclosures, which have a window for viewing visible blade position, are available with 30 to 400 A ratings. The window also allows viewing of indicating fuses for 30 to 200 A fusible switches.

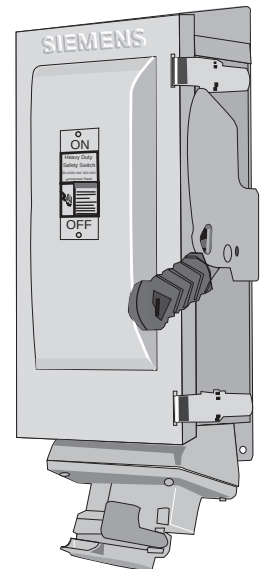
Siemens heavy duty safety switches are also available in 30 to 200 A ratings with 316 grade stainless steel Type 4/4X enclosures with or without a viewing window. These enclosures are more corrosion resistant than the standard 304 grade stainless steel enclosures. Type 316 stainless steel enclosures are especially suited for environments containing a high level of chlorine or other chemicals commonly encountered in marine, waste management, food and beverage, petrochemical, and mining applications.

Interlock Receptacle Safety Switches

Interlock receptacle safety switches

provide a receptacle for powering heavy-duty portable equipment such as refrigerated trucks, welders, and other portable electric tools. These switches are fitted with a Crouse-Hinds Arktite or similar receptacle which is interlocked to prevent insertion or removal of the plug when the switch is in the "on" position. The Crouse-Hinds receptacle switch requires a Crouse-Hinds 4-wire, 3-pole, style 2, grounded APJ plug.

Interlock receptacle safety switches are rated for 30, 60, and 100 amperes. These switches are available with Type 12 or 4/4X enclosures.



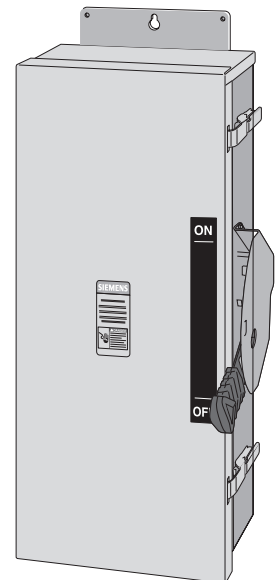
Receptacle

4-Pole and 6-Pole Safety Switches

4-pole and 6-pole heavy-duty fusible and non-fusible safety switches

are available with current ratings of 30 to 200 amperes. 4-pole switches are available with either a Type 1 or Type 12/3R enclosure. 6-pole switches are available with either a Type 12/3R enclosure or Type 4X stainless steel enclosure.

These switches are commonly used as a disconnecting means for 2-speed, 2-winding motors. A 4-pole switch is also used in 3-phase, 4-wire circuits when a switching neutral is required.

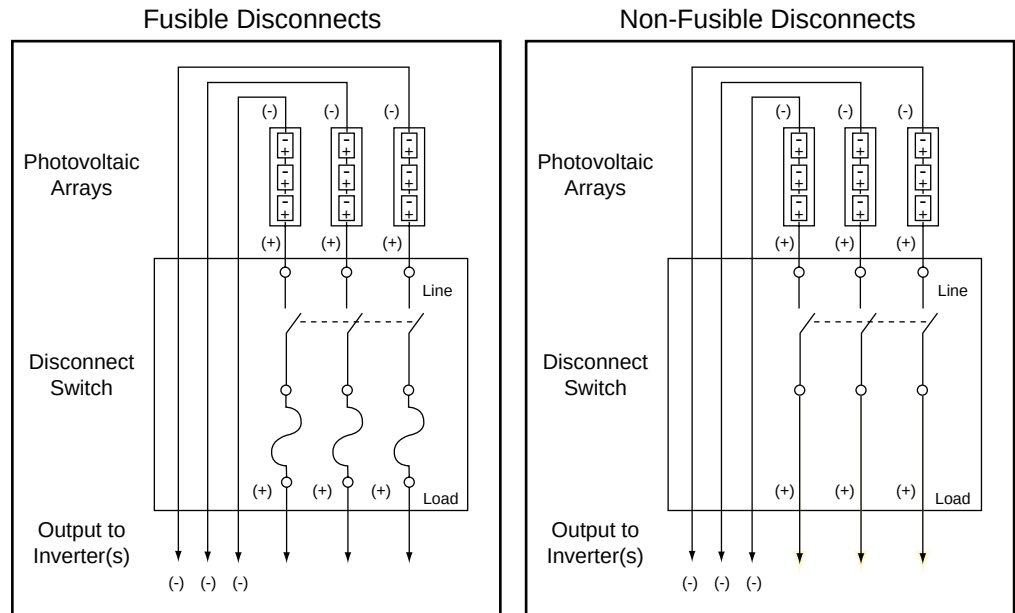


Enclosed Photovoltaic Disconnect Switches

Siemens enclosed photovoltaic disconnect switches have passed the extremely rigorous testing required by UL 1741 for photovoltaic disconnects with three separate 600 VDC (max.) circuits connected to a single, three-pole switch. Use of these disconnects reduces system cost in comparison to systems which provide separate disconnects for each circuit.

A very hot arc is generated by the interruption of a 600 VDC circuit when under load. This damages conventional safety switch contacts and insulating material after a small number of operations unless the poles are connected in series to spread this destructive energy over at least two sets of contacts.

In contrast, Siemens photovoltaic disconnect switches have powerful magnets in their line base which are strategically located and specifically aligned to disperse this energy and to very quickly extinguish the arc. The result is a line of disconnects that performs at a level far beyond that of any conventional safety switch.



Note: Siemens enclosed photovoltaic disconnect switches are for use only on negative ground systems.

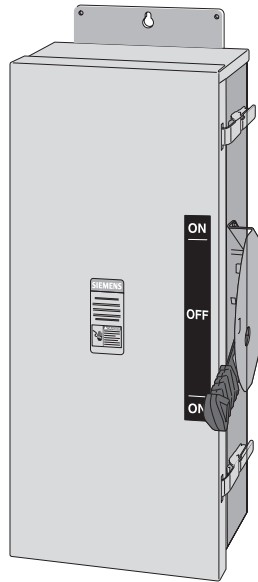
These switches are available with Type 1 or Type 3R enclosures in fusible and non-fusible versions with continuous current rating from 30 to 100 A. Line and load lugs accept larger conductors than required by UL. This allows larger cables to be used to reduce voltage drop. A unique cover design features a rolled out front flange that prevents cuts and scrapes to conductor installation and the installer's hands.

Additional standard features include:

- Factory-installed ground bar
- Door labeling as required by *NEC*® Article 690
- Two-point and three-point mounting provisions are provided
- Large top, bottom, and side gutters make wiring easier

Double Throw Switches

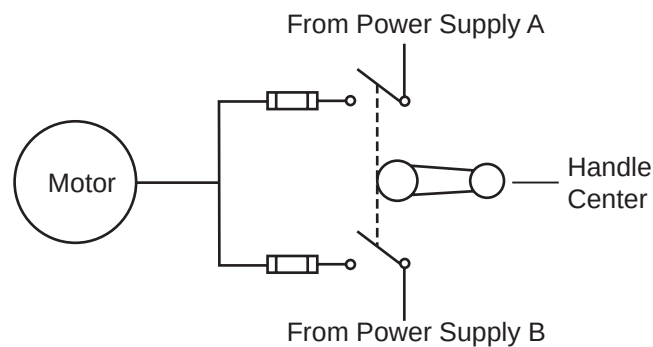
Heavy duty, double throw, non-fusible switches are available with current ratings of 30 to 1200 amps. General duty, double throw, non-fusible switches are available with current ratings of 100 or 200 amperes. Most products are available with a Type 1 or Type 12/3R enclosure. A few versions are available with a Type 12/3R or Type 4X enclosure.



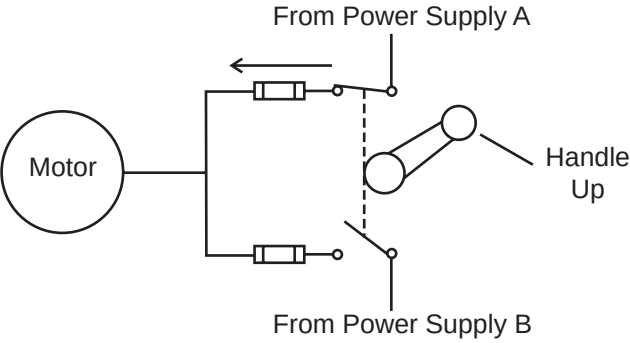
Double Throw Switch Application

Double throw switches are used to connect a single power source to either of two loads or to transfer loads from one power source to another.

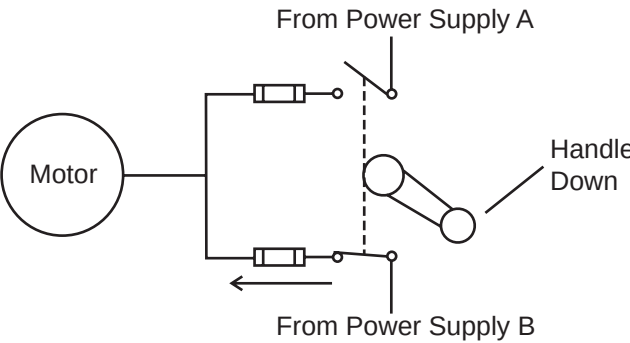
For example, a critical piece of equipment often needs a back-up power supply in case the main power supply fails or needs maintenance. In the following example, a motor can be connected through a **double throw switch** to power supply A or power supply B. When the handle is in the center position, the switch is off and no power flows to the motor.



Moving the handle to the up position connects the motor to power supply A.



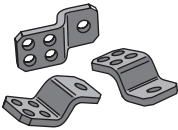
Moving the handle to the down position connects the motor to power supply B.



Safety Switch Accessories

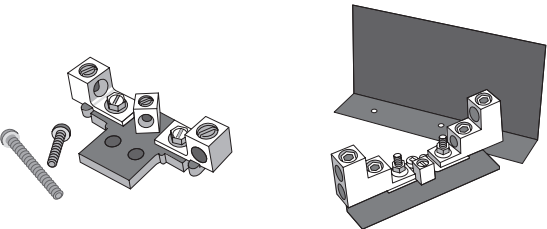
A full range of **accessories** is available for Siemens VBII safety switches. Some of these are shown below.

Both general duty and heavy duty switches are field-convertible to accept Class J or Class T fuses.



HT63
Class T Fuse Adapter Kit

Standard neutral kits can be field installed in both general duty and heavy duty safety switches. UL listed 200% Neutrals are available on 100-600A heavy duty switches.



HN612

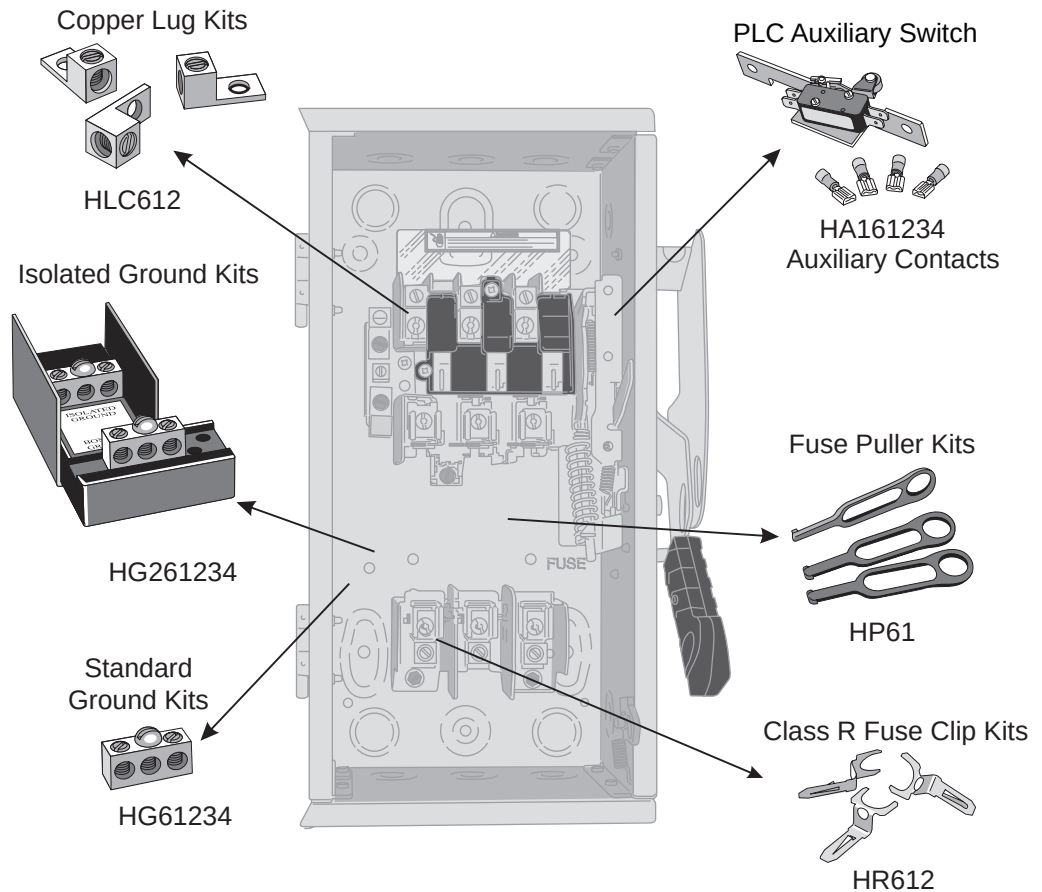
HN264

The multiple padlock accessory is a tamper-proof device to provide for multiple padlocking to meet OSHA or plant requirements.



SL0420
Multiple Padlock Accessory

The following illustration shows some of the other accessories available for general and heavy duty safety switches.



Heavy duty switches are UL approved to accept field installed copper lug kits. Equipment ground kits are available for all general duty and heavy duty switches. They come standard in Type 4/4X and Type 12 switches and can be installed in the field in Type 1 and Type 3R switches.

Isolated ground kits are also available for 30 to 600 A heavy duty switches. Some circuits with a high degree of computer or other electronic loading require an isolated ground to prevent interference from the building ground and neutral lines.

Auxiliary contacts are available only for heavy duty switches. They come with one normally open and one normally closed or two normally open and two normally closed contacts. A PLC auxiliary switch is also available for 30 to 200 A safety switches. It has very low contact resistance, which is compatible with the low voltages and currents typically found in PLC circuits.

Fuse puller kits can be installed in 30 to 100A heavy duty switches in the field.

Class R fuse clips are used to prevent the installation of non-current-limiting Class H or Class K fuses. All general and 30 to 600 A heavy duty switches are field convertible to accept Class R fuse clip kits.

Catalog Numbers

Each type of safety switch has a catalog number. The catalog number provides a description of the safety switch. There are eight parts to the catalog number for a Siemens VBII Safety Switch. The following figure illustrates a typical catalog number.

Catalog Number	Part 1	Part 2	Part 3	Part 4	Part 5	Part 6	Part 7	Part 8
HF364NRCU=	H	F	3	6	4	N	R	CU

Part 1

Part 1 indicates the switch type. There are five types available: General Duty 10 kA interrupting rating (Plug Fused and 60 A max. Non-Fused), General Duty, Heavy Duty, Heavy Duty Double Throw, and General Duty Double Throw.

From following table shows that the example switch, type H, is a heavy duty switch.

Designator	Switch Type
L	General Duty 10k AIC Max.
G	General Duty
H	Heavy Duty
DT	Heavy Duty Double Throw
DTG	General Duty Double Throw

Part 2

Part 2 indicates whether the switch is fused or non-fused. **F** designates a fused switch, and **NF** designates a non-fused switch. For this example, the switch is fused.

Part 3

Part 3 of the catalog number indicates the number of poles. Siemens VBII safety switches can be provided with 1, 2, 3, 4, or 6 poles. A neutral, if required, is not included in the number of poles. The example catalog number calls for a 3-pole safety switch.

Part 4

Part 4 of the catalog number indicates the voltage rating. The example catalog number indicates a safety switch with a maximum voltage rating of 600 volts.

Designator	Voltage
1	120V or 120/240V
2	240V
6	600V

Part 5

Part 5 of the catalog number refers to the switch's current rating. The example indicates a safety switch with a 200 ampere rating.

Designator	Amperes
1	30A
2	60A
3	100A
4	200A
5	400A
6	600A
7	800A
8	1200A

Part 6

Part 6 of the catalog number indicates whether or not a neutral is included with the switch. If no neutral is needed, part 6 of the catalog number is simply omitted. If a neutral is needed, an **N** is added to the catalog number, as in the example.

Part 7

Part 7 of the catalog number indicates the type of enclosure. The example catalog number indicates a safety switch in a NEMA Type 3R outdoor enclosure.

Designator	Enclosure Type
Omit	Type 1, Indoor
R	Type 3R, Outdoor
S	Type 4/4X, 304 Grade Stainless Steel
SS	Type 4/4X, 316 Grade Stainless Steel
X	Type 4/4X, Non-Metallic
J	Type 12, Industrial

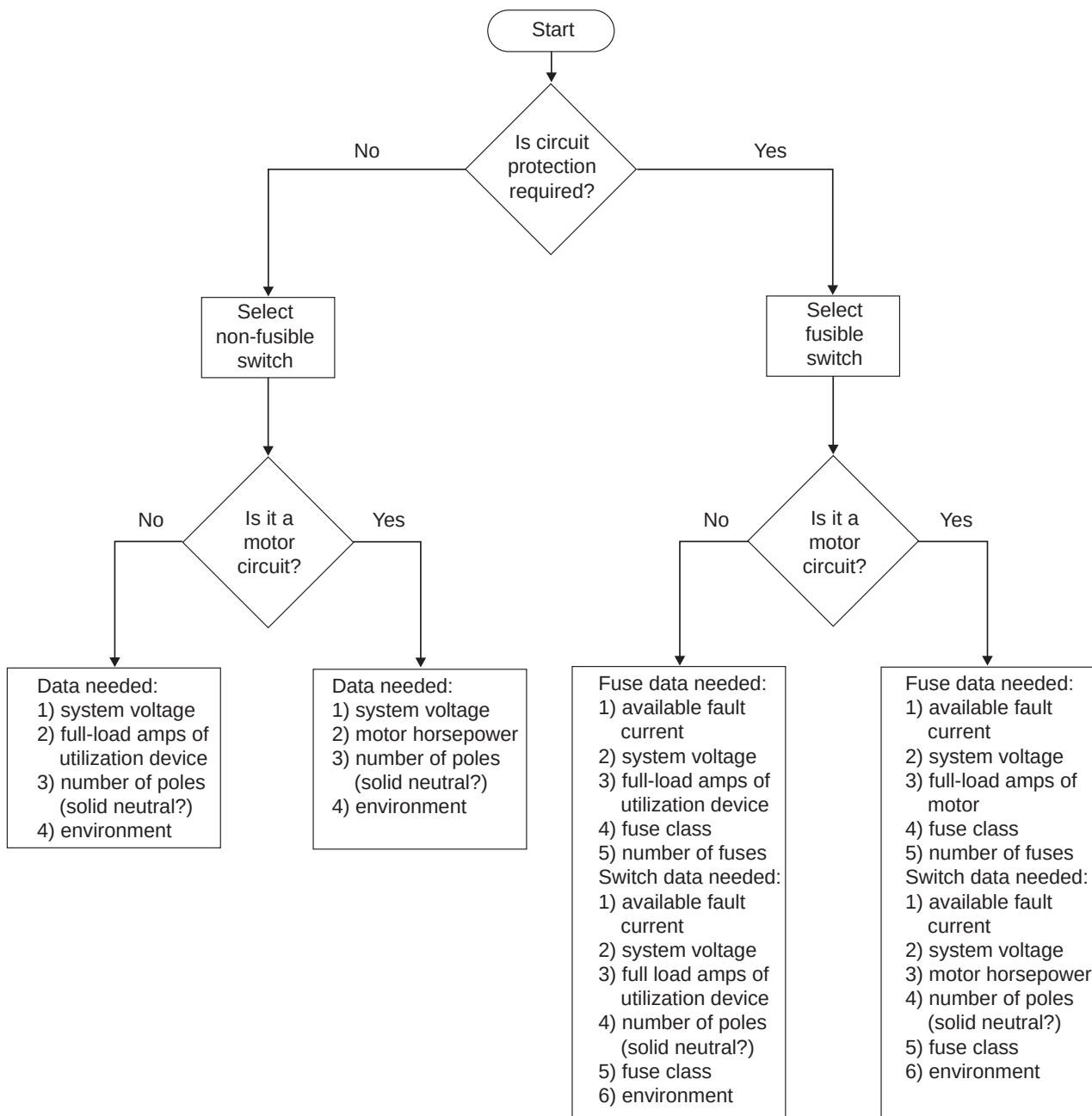
Part 8

Part 8 of the catalog number is for special applications. The following table lists the possible applications. For example, **CU** indicates factory-installed copper wire grips, as in the example catalog number.

Designator	Special Applications:
CH	Crouse-Hinds Receptacle
CJ	Factory J Fuse Spacings
CR	Class R Clips Installed
CU	Copper Wire Grips Installed
G	Factory-Installed Ground Bar
PN	Pyle-National Receptacle
PV	Photovoltaic Disconnect Switch
W	Viewing Window

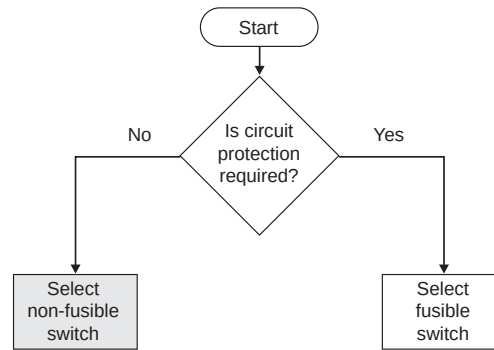
Selecting Safety Switches

While selecting a safety switch is not difficult, flow charts can help to make it even easier. The following flow chart can be used to make key decisions in the selection of a safety switch.



Selecting a Non-Fusible Switch

Is circuit protection required? If circuit protection is not required a non-fusible switch would be selected.



Non-Fusible Switch not Used on a Motor Circuit

If a non-fusible switch is selected, is it for a motor circuit? If the switch is not used in a motor circuit, the following information must be known:

- 1) System voltage: 120 VAC, 240 VAC, 480 VAC, 600 VAC, 250 VDC, 600 VDC
- 2) Full-load amperes of the device to be used on the switch
- 3) The number of poles required, and if a neutral is needed
- 4) The environment (enclosure type)

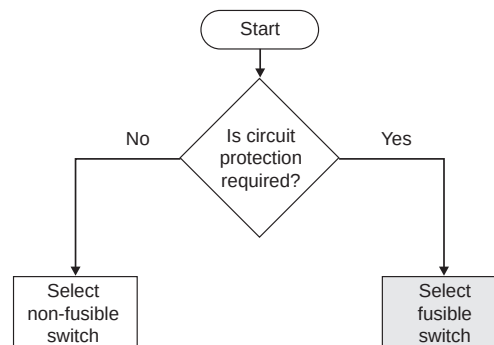
Non-Fusible Switch Used in a Motor Circuit

If the switch is used in a motor circuit, the same data is required, except that motor horsepower replaces full-load current.

- 1) System voltage
- 2) Motor horsepower
- 3) The number of poles required, and if a neutral is needed
- 4) The environment (enclosure type)

Selecting a Fusible Switch

If circuit protection is required, a fusible switch would be selected.



Fusible Switch not Used on a Motor Circuit

If a fusible switch is selected, is it for a motor circuit? If not, the following information must be known to select fuses:

- 1) Available fault current
- 2) System voltage
- 3) Full-load amperes of the device to be used on the switch
- 4) Fuse class
- 5) Number of lines to be fused

The following must be known to select a switch:

- 1) Available fault current
- 2) System voltage
- 3) Full-load amperes of the device to be used on the switch
- 4) Number of poles, and if a neutral is needed
- 5) Fuse class
- 6) Environment (enclosure type)

Fusible Switch Used on a Motor Circuit

If the switch is used on a motor circuit, the following information must be known to select a fuse:

- 1) Available fault current
- 2) System voltage
- 3) Full-load amperes required by the motor
- 4) Fuse class
- 5) Number of lines to be fused

The following must be known to select a switch:

- 1) Available fault current
- 2) System voltage
- 3) Motor horsepower
- 4) Number of poles, and if a neutral is needed
- 5) Fuse class
- 6) Environment (enclosure type)

Example of Selecting a Non-Fusible Safety Switch

In the following example, a safety switch needs to be provided for an application that does not require circuit protection. The continuous load is 45 amperes. There is no non-continuous load. It is not a motor circuit. The system voltage is 240 VAC, 3-phase, 3-wire (without neutral). The environment is indoors, and there are no unusual conditions such as dust or liquids.

According to *NEC*® Article 210.19, all conductors (including the safety switch) in branch circuits that are not supplying only motor loads must be capable of carrying 125% of the continuous load (maximum current lasting 3 hours or more) plus 100% of any non-continuous load. In this example, a switch must be selected that can carry 56 amperes.

$$\begin{array}{r} 45 \text{ amperes} \\ \times 1.25 \\ \hline 56 \text{ amperes} \end{array}$$

Knowing that the switch will be used indoors, with no unusual conditions, a Type 1 enclosure can be selected. The other requirements can be met with a general duty switch. Referring to the General Duty Safety Switches section of the Speedfax catalog, the first 240 volt, 3-pole, non-fusible switch that will handle 56 amperes is a 60 amp switch. The catalog number is GNF322.

System	Ampere Rating	Indoor - Type 1			Outdoor - Type 3R		
		Catalog Number	List Price \$	Ship Wt. Std. Pkg.	Catalog Number	List Price \$	Ship Wt. Std. Pkg.

240 Volt Non-Fusible

2-Pole or 3-Pole

	30	GNF321			GNF321R		
	60	GNF322			GNF322R		
	100	GNF323			GNF323R		
	200	GNF324			GNF324R		
	400	GNF325			Use 600V Switch - HNF365R Use 600V Switch - HNF366R		
	600	GNF326					

Example of Selecting a Fusible Safety Switch

In the following example, a safety switch needs to be provided for an application that does require circuit protection. This application has a 480 VAC, 3-phase, 75 HP, NEMA design B, energy-efficient motor, that does not need a neutral connection. The customer has specified RK5 time-delay fuses. The switch will be located indoors with no unusual service conditions.

The application requirements for this example dictate selection of a heavy duty, 600 volt, fusible switch. On the appropriate Speedfax page, locate the enclosure type, Indoor — Type 1.

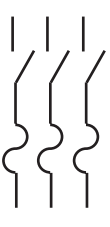
Next, find the 600 volt, fusible, 3-pole, 3-fuse table. In the 480 VAC, 3-phase, 3-wire section of this table, select a switch with a horsepower equal rating in the maximum (Max.) column that is equal to or greater than 75. The maximum column must be used because the customer selected time delay fuses. (Had non-time delay fuses been specified, the standard horsepower column would be used.) In this example, 125 HP is the first rating meeting the 75 HP requirement. Reading to the left, the catalog number under Indoor - Type 1 is HF364. Also note that this switch has an ampere rating of 200.

Because a Class R fuse is required for this application, a Class R fuse clip kit is also required. This can be found in the accessory section of the Speedfax. In this example, the fuse kit catalog number is HR64.

System	Ampere Rating	Indoor - Type 1			Horsepower Ratings			
		Catalog Number	List Price \$	Ship Wt. Std. Pkg.	480 VAC			
					1 Phase, 2 Wire		3 Phase, 3 Wire	
					Std.	Max.	Std.	Max.

600 Volt Fusible

3-Pole, 3-Fuse

	30	HF361			3	7½	5	15
	60	HF362			5	20	15	30
	100	HF363			10	30	25	60
	200	HF364			25	50	50	125
	400	HF365					100	250
	600	HF366					150	400
	800	HF367					200	500
	1200	HF368					200	500

Selecting a Fuse

Section 430.6 of the *NEC*® requires that, where the current rating of a motor is used to determine the ampacity of conductors or ampere ratings of switches, branch-circuit overcurrent devices, etc., the values given in Tables 430.247 through 430.250 must be used instead of the actual motor nameplate current rating. According to *NEC*® Table 430.250, a 75 HP, 460 VAC motor has a full-load current of 96 amperes.

Table 430.250 Full-Load Current, Three-Phase Alternating-Current Motors

The following values of full-load currents are typical for a motor running at speeds usual for belted motors and motors with normal torque characteristics.

The voltages listed are rated motor voltages. The currents listed shall be permitted for system voltage ranges of 110 to 120, 220 to 240, 440 to 480, and 550 to 600 volts.

Horsepower	Induction-Type Squirrel Cage and Wound Rotor (Amperes)						
	115 Volts	200 Volts	208 Volts	230 Volts	460 Volts	575 Volts	2300 Volts
½	4.4	2.5	2.4	2.2	1.1	0.9	—
¾	6.4	3.7	3.5	3.2	1.6	1.3	—
1	8.4	4.8	4.6	4.2	2.1	1.7	—
1½	12.0	6.9	6.6	6.0	3.0	2.4	—
2	13.6	7.8	7.5	6.8	3.4	2.7	—
3	-	11.0	10.6	9.6	4.8	3.9	—
5	-	15.5	16.7	15.2	7.6	6.1	—
7½	-	25.3	24.2	22	11	9	—
10	-	32.2	30.8	28	14	11	—
15	-	48.3	46.2	42	21	17	—
20	-	62.1	59.4	54	27	22	—
25	-	78.2	74.8	68	34	27	—
30	-	92	88	80	40	32	—
40	-	120	114	104	52	41	—
50	-	150	143	130	65	52	—
60	-	177	169	154	77	62	16
75	-	221	211	192	96	77	20
100	-	285	273	248	124	99	26
125	-	359	343	312	156	125	31
150	-	414	396	360	180	144	37
200	-	552	528	480	240	192	49
250	-	-	-	-	302	242	60
300	-	-	-	-	361	289	72
350	-	-	-	-	414	336	83
400	-	-	-	-	477	382	95
450	-	-	-	-	515	412	103
500	-	-	-	-	590	472	118

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Table 430.52 of the *NEC*® is provided to help select a fuse that will not open while starting a motor and will still provide adequate overcurrent protection. According to this table, the *NEC*® requires that the ampere rating of an AC motor protected by a time-delay fuse be multiplied by 1.75.

Table 430.52 Maximum Rating or Setting of Motor Branch-Circuit Short-Circuit and Ground-Fault Protective Devices

Type of Motor	Percentage of Full-Load Current			
	Nontime Delay Fuse	Dual Element (Time-Delay) Fuse	Instantaneous Trip Breaker	Inverse Time Breaker
Single-phase motors	300	175	800	250
AC polyphase motors other than wound-rotor	300	175	800	250
Squirrel cage - other than Design B energy-efficient	300	175	800	250
Design B energy-efficient	300	175	1100	250
Synchronous	300	175	800	250
Wound rotor	150	150	800	150
Direct current (constant voltage)	150	150	250	150

Multiplying the motor rating of 96 amperes times 1.75 results in a fuse size of 168 amperes. Since this is a non-standard fuse size, the next standard fuse size of 175 amperes should be selected.

96 amperes	Full-Load Motor Current
<u>x 1.75</u>	NEC® Requirement
168 amperes	Fuse Rating

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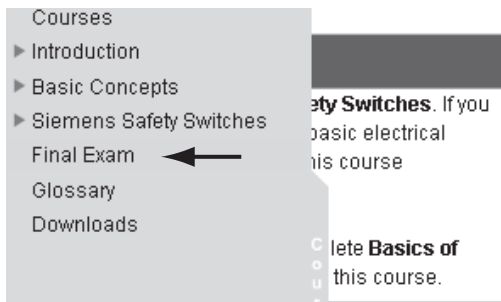
Review Answers

Final Exam

Before taking the final exam, it is recommended that you delete the temporary internet files from your computer's web browser. For most versions of **Internet Explorer**, you can do this by selecting **Internet Options** from the **Tools** menu and then clicking on the **Delete Files** button. If you do not perform this step, you may see a score of 0% after you submit your exam for grading.

The final exam for this course is available online at **<http://www.usa.siemens.com/step>**. This web page provides links to all our quickSTEP online courses. To complete the final exam for this course, click on the **Basics of Safety Switches** link.

Next, move your mouse over to the left so that the navigation bar pops out and select the **Final Exam** link. The final exam page will appear.



After you complete the final exam, click on the **Grade the Exam** button at the bottom of the page. Your score on the exam will be displayed along with the questions that you missed.

If you score 70% or better on the exam, you will be given two options for displaying and printing a certificate of completion. The **Print Certificate** option allows you to display and print the certificate without saving your score in our database and the **Save Score** option allows you to save your score and display and print your certificate. **The Save Score option is primarily intended for use by our distributors and Siemens employees.**